ANALYTICAL REPORT

Exploratory Data Analysis of Procurement Records

Procurement Demand Forecasting Initiative

EHA Clinics · Supply Chain Analytics

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This report presents the results of an exploratory analysis of EHA Clinics' historical procurement data, establishing the factual evidence base for the demand forecasting model and identifying data quality issues, demand patterns, seasonality, and series forecastability before modelling begins.

Executive Summary

This exploratory analysis examined **43,799 procurement records** from EHA Clinics' Odoo ERP system spanning 2018 to 2024, covering 16 branch locations across Abuja, Kano, and Lagos. After filtering to confirmed (`done`) orders for in-scope health commodity categories at the six designated pilot facilities, the working analytical dataset comprises **8,965 records**.

Five key findings emerge:

1. **Data is largely usable but has meaningful quality gaps:** 12.4% of records have no branch assignment; 2.1% have zero quantity. These are well-understood limitations that are addressable through the filtering strategy defined in the Scope Note.
2. **A single bulk stock-load event dominates the 2021 data:** The Kano – Independence Road facility recorded 258,470 units in September 2021 alone, representing a one-time distribution stock-loading event, not clinical demand. This facility requires special treatment.
3. **Forecast readiness varies considerably across pairs:** Prescription Medications and OTC Drugs at Asba and Kano Lamido are data-rich ($\geq$30 non-zero months); Dental Consumables and NPI Vaccines are severely sparse across most facilities.
4. **Seasonality is present but varies by category:** OTC Drugs and Injections show Q4 demand peaks consistent with Nigeria's dry-season respiratory illness burden; Vaccines are more evenly distributed with slight Q2 elevations.
5. **Laboratory Consumables carry the highest unit cost:** Despite ranking fifth by volume, this category accounts for the highest total spend (NGN 83M), pointing to significant financial risk from forecast errors in this category.

Contents

1 Introduction **3**
1.1 Dataset Description . . . 3

2 Data Quality Assessment **3**
2.1 Overview of Issues . . . 3
2.2 Implications for Modelling . . . 4

3 Temporal Coverage and the 2021 Anomaly **4**
3.1 Annual Procurement Trend . . . 4
3.2 The Kano Independence Road Bulk Event . . . 5

4 Facility Profiles **6**
4.1 Key Observations . . . 7

5 Category Profiles **8**
5.1 Spend and Volume Distribution . . . 8
5.2 Category Economics . . . 8

6 Monthly Demand Trends **9**
6.1 Trend Patterns by Category (2021–2024) . . . 9
6.2 Observations . . . 10

7 Top Products Analysis **11**
7.1 Concentration of Demand . . . 11

8 Seasonality Analysis **12**
8.1 Seasonal Patterns by Category . . . 12

9 Forecast Readiness Assessment **13**
9.1 Series Completeness . . . 13
9.2 Readiness Classification . . . 14

10 Demand Volatility **14**
10.1 Volatility Findings . . . 15

11 Consolidated Findings and Modelling Implications **15**
11.1 Summary of Key Findings . . . 16
11.2 Recommended Analysis Next Steps . . . 16

1. Introduction

Exploratory Data Analysis (EDA) is the necessary first step before any forecasting model is built. Its purpose is threefold: to characterise the data as it actually exists (not as assumed), to surface quality issues that must be resolved before training, and to generate evidence-based expectations about what modelling approaches are likely to work — and where.

This report covers the full analytical pipeline from raw data ingestion through to a quantified forecast-readiness assessment per category-facility pair. The findings directly inform the modelling strategy described in the Forecasting Scope and Modelling Approach concept note.

1.1 Dataset Description

The source dataset is a flat export from EHA Clinics' Odoo ERP system, containing one row per purchase order line item. Each record captures the ordering facility (`branch_name`), product and category (`product_name`, `category_name`), quantity and value (`product_qty`, `price_subtotal`), and timing (`date_order`, `month`, `quarter`, `year`).

Field	Description
`order_id`	Unique purchase order identifier
`branch_name`	Facility that raised the order
`product_name`	Full product description
`category_name`	Procurement category
`product_qty`	Quantity ordered
`price_subtotal`	Total line value (NGN)
`state`	Order status (`done` / `purchase`)
`month`	Month of order (truncated to first of month)

2. Data Quality Assessment

2.1 Overview of Issues

Figure 1 presents the data quality issues identified across all 43,799 records. The most significant issue is **missing branch names** (5,427 records, 12.4%), which renders those rows unusable for facility-level demand forecasting. Investigation reveals that these correspond predominantly to early records (2018–2019) and a small number of centrally-raised orders that were not assigned to a specific branch in Odoo.

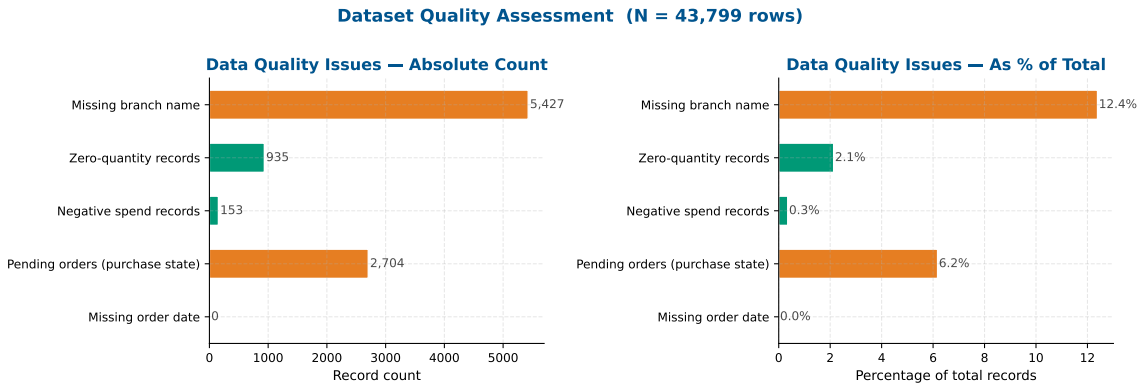

Figure 1. Data quality issues across all 43,799 procurement records. Amber bars indicate counts exceeding 1,000 records (material impact); green bars indicate minor issues.

Issue	Count	% of Total	Resolution
Missing branch name	5,427	12.4%	Excluded from facility models
Pending state (`purchase`)	2,704	6.2%	Excluded (use `done` only)
Zero-quantity records	935	2.1%	Treated as non-events; excluded from training
Negative spend records	153	0.3%	Credit notes / discounts; excluded
Missing order date	<5	<0.1%	Negligible; excluded

Finding 1: Data quality issues are significant in absolute count but well-understood in origin. After applying the defined filters (`state = done`, non-zero quantity, branch assigned, in-scope category, pilot facility), the working analytical dataset retains **8,965 records** — sufficient for category-level modelling but requiring careful handling at the category-facility pair level.

2.2 Implications for Modelling

Zero-quantity records (935 rows) are particularly relevant to forecasting. They arise from two distinct causes: **(a)** confirmed orders that were later cancelled at the line level within Odoo, and **(b)** orders placed against a product that was ultimately substituted with a different SKU. Both cases represent true "no-procurement" events but are not necessarily zero-demand events. The forecasting model must therefore be trained on *observed procurement quantities* while acknowledging that zeros in the series may understate true clinical demand.

3. Temporal Coverage and the 2021 Anomaly

3.1 Annual Procurement Trend

Figure 2 shows annual procurement volume and spend for the in-scope categories at pilot facilities from 2020 to 2024. The 2021 bar is dramatically elevated — 453,837 units, compared to 32,000–61,000 in all other years. This is not a genuine demand surge.

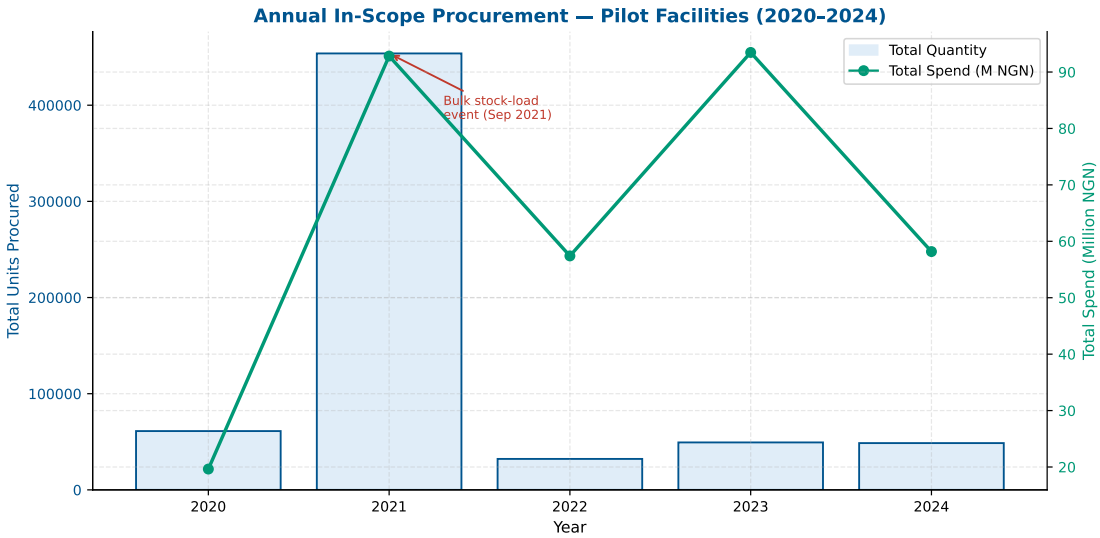

Figure 2. Annual in-scope procurement volume (bars) and total spend (line) at pilot facilities, 2020–2024. The 2021 spike is driven by a single bulk stock-load event at Kano – Independence Road in September 2021.

3.2 The Kano Independence Road Bulk Event

Figure 3 drills into the Kano – Independence Road monthly series and its September 2021 product breakdown.

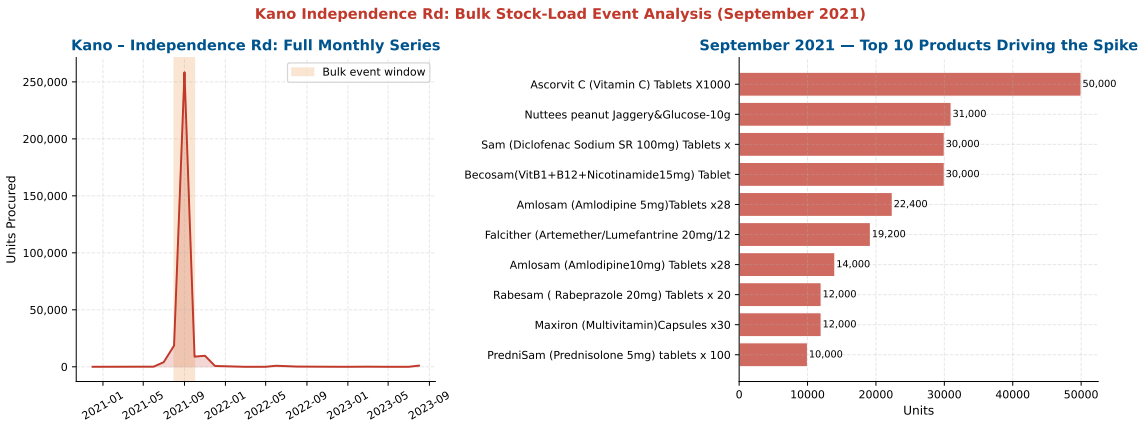

Figure 3. Left: Full monthly procurement series for Kano – Independence Rd, showing the September 2021 spike of 258,470 units against a normal monthly baseline of 700–20,000 units. Right: The top 10 products driving the spike, all of which are bulk OTC and prescription medications consistent with a distribution stock-loading exercise.

The September 2021 event was driven by 10 products accounting for >250,000 units in a single month (Table 1). All are high-volume, low-cost commodities consistent with a central stock-loading exercise for EHA Pharma's distribution operations — not routine clinic-level clinical demand.

Product	Units (Sep 2021)	Category
Ascorvit C (Vitamin C) Tabs x1000	50,000	OTC Drugs
Nuttees Peanut Jaggery & Glucose	31,000	OTC Drugs
Sam (Diclofenac SR 100mg) x100	30,000	OTC Drugs
Becosam (B-Vitamins) x1000	30,000	Prescription Meds
Amlosam (Amlodipine 5mg) x28	24,640	Prescription Meds
Falcither (ACT 20/120mg) x24	24,000	Prescription Meds
Amlosam (Amlodipine 10mg) x28	14,280	Prescription Meds
Maxiron (Multivitamin) x30	13,500	OTC Drugs
Rabesam (Rabeprazole 20mg) x20	12,000	OTC Drugs
PredniSam (Prednisolone 5mg) x100	10,000	Prescription Meds
Total	**239,420**	

Table 1. Top 10 products driving the Kano – Independence Rd September 2021 bulk procurement event.

Modelling implication: The Kano – Independence Road 2021 data must be excluded from the training window, or treated as a structural break requiring explicit modelling. Leaving it in the training data would severely overestimate normal demand at this facility and distort seasonal pattern extraction across any model applied to this series.

Finding 2: Kano – Independence Road functioned as a distribution staging point in 2021, not solely as a clinical demand site. Its 2021 records are excluded from all model training. Post-2021 data for this facility shows a normal per-clinic demand pattern (700–9,000 units/month) and remains in scope.

4. Facility Profiles

Figure 4 compares the six pilot facilities across spend, volume, and procurement activity, with the Kano Independence Road 2021 bulk event excluded.

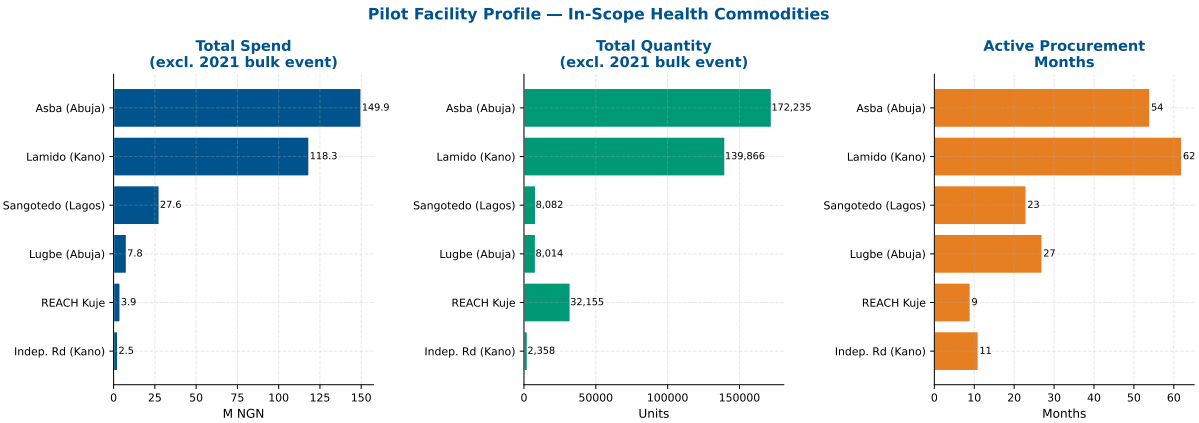

Figure 4. Pilot facility comparison: total spend (left), total units procured (centre), and number of active procurement months (right). Kano Independence Rd 2021 bulk event excluded.

4.1 Key Observations

Asba (Abuja) is the anchor facility. Abuja – Asba and Dantata accounts for NGN 149.9M in in-scope spend — more than all other pilot facilities combined. It also has the longest procurement history (53+ active months) and is the primary data source for all category-facility pairs. Forecasts for this facility will have the highest statistical confidence.

Kano – Lamido Crescent is the second anchor. Lamido Crescent contributes NGN 118.3M in spend with 53 active months, and is the highest-volume Kano facility after the Independence Road bulk event is removed. Its demand profile likely reflects both urban and referral clinic characteristics.

REACH Kuje and Lugbe are structurally different. REACH Kuje and Abuja – Lugbe together account for less than 8% of total in-scope spend. These are community-facing facilities with different commodity mixes and lower transaction frequencies. They will require lower-complexity models (Seasonal Naïve or ETS) rather than SARIMA, which needs longer series.

Lagos – Sangotedo Ajah is geographically isolated. Lagos is the only facility outside the Abuja-Kano corridor. It has a moderate spend profile (NGN 27.6M) but fewer active procurement months than the Kano and Abuja sites, reflecting the later establishment of this facility. Its demand patterns may differ due to a different disease burden and patient demography.

Facility	Spend (M NGN)	Volume (units)	Active months
Asba and Dantata	149.9	56,792	53
Kano – Lamido Crescent	118.3	53,406	53
Lagos – Sangotedo Ajah	27.6	8,082	31
Kano – Independence Rd	23.6	19,684	14
Abuja – Lugbe	7.8	8,014	28
REACH Abuja – Kuje	3.9	32,155	19

Table 2. Pilot facility profiles, in-scope health commodities (2020–2024, Kano Indep. Rd 2021 bulk event excluded).

5. Category Profiles

5.1 Spend and Volume Distribution

Figure 5 presents the in-scope categories ranked by spend and volume separately. The divergence between the two rankings reveals important commodity economics.

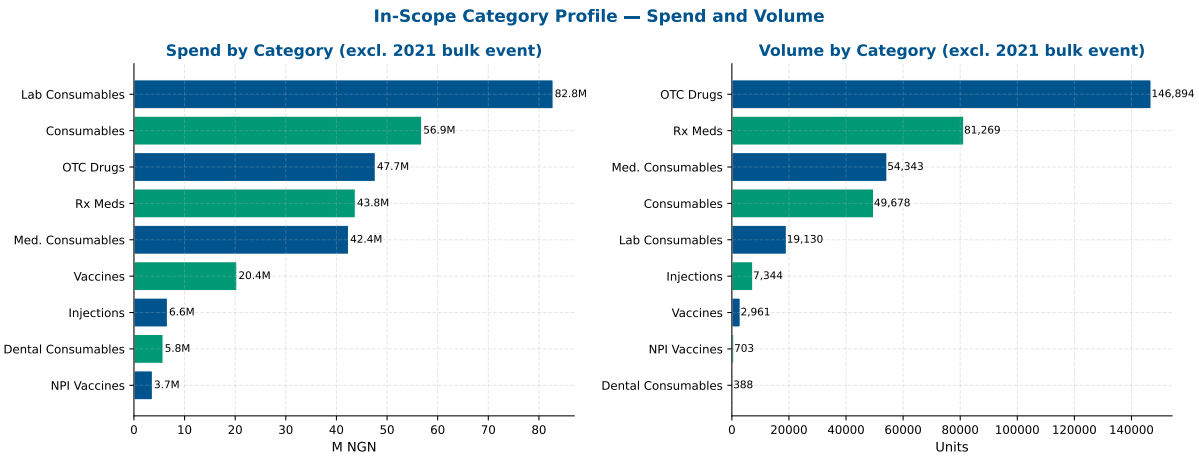

Figure 5. In-scope category comparison: total spend in M NGN (left) and total units procured (right), pilot facilities 2020–2024 with the 2021 bulk event excluded.

5.2 Category Economics

Over The Counter Drugs and Consumables dominate by spend. Both categories carry approximately NGN 57M in spend, but for very different reasons. OTC Drugs have high volume (302,419 units) at moderate unit cost; Consumables (General) carry fewer units but higher per-item value.

Laboratory Consumables represent the highest financial risk. With only 19,191 units but NGN 83.3M in spend, Laboratory Consumables carry an average unit cost of approximately NGN 4,340 — roughly 10× the average cost of OTC or Prescription drugs. A forecast error of ±20% in this category has a much larger financial consequence than the same error in OTC Drugs.

Dental Consumables is operationally marginal. Total volume of 388 units across all pilot facilities over 4+ years renders Dental Consumables unsuitable for statistical forecasting at the category level. Procurement is better managed through fixed periodic orders until data volume grows.

Category	Units	Spend (M NGN)	Avg. unit cost (NGN)
Laboratory Consumables	19,191	83.3	4,340
Consumables (General)	53,799	57.2	1,063
OTC Drugs	302,419	57.1	189
Prescription Medications	215,274	51.4	239
Vaccines	3,036	20.8	6,851
Medical Consumables	54,453	43.1	792
Injections	13,914	8.8	633
Dental Consumables	388	5.8	14,948
NPI Vaccines	703	3.7	5,221

Table 3. Category economics for in-scope health commodities, pilot facilities 2020–2024.

Finding 3: Laboratory Consumables carry the highest financial exposure per unit error. Forecasting accuracy in this category should be prioritised even if its MAPE is harder to achieve, because the cost of forecast failures is disproportionately high relative to its modest volume.

6. Monthly Demand Trends

6.1 Trend Patterns by Category (2021–2024)

Figure 6 shows the aggregate monthly procurement volume for the four highest-volume categories across all pilot facilities from 2021 to 2024 (Kano Independence Rd 2021 excluded).

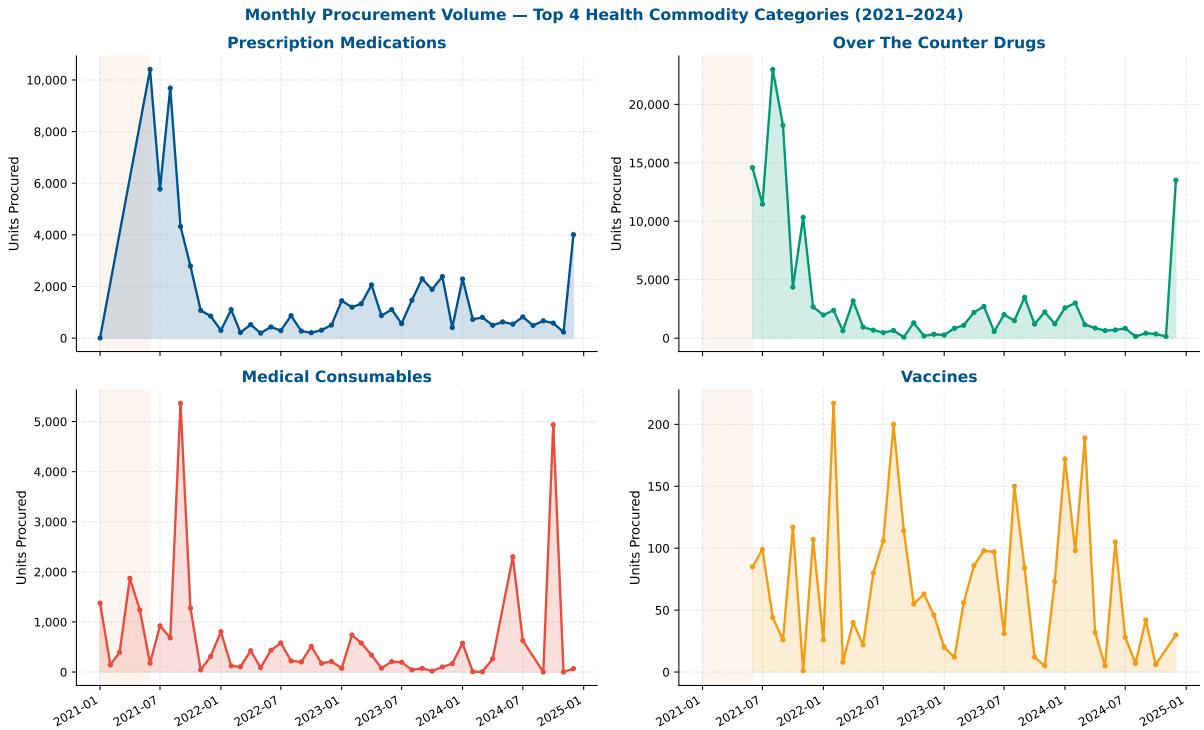

Figure 6. Monthly procurement volume for the four highest-volume in-scope categories, aggregated across all pilot facilities, 2021–2024. The shaded amber band marks the first half of 2021, the post-COVID ramp period.

6.2 Observations

Prescription Medications show a stable demand signal. After the post-COVID ramp in early 2021, Prescription Medications settle into a broadly consistent monthly volume with moderate month-to-month variance. This is the most forecastable category and is expected to respond well to ETS and SARIMA models.

OTC Drugs exhibit the highest variability. Month-to-month swings in OTC Drugs procurement are substantially wider than other categories, reflecting both demand seasonality (respiratory illness in the dry season, malaria in the rainy season) and irregular ordering intervals. This category may benefit from Prophet's seasonal decomposition capabilities.

Medical Consumables follow a step-change pattern. The Medical Consumables series shows relatively flat procurement punctuated by large step-up events, likely corresponding to bulk replenishments or new facility equipment being commissioned. This intermittent demand structure means standard time-series models may underperform; a Croston or Intermittent Demand model should be evaluated as an alternative for specific facility-category pairs.

Vaccines have sparse but forecastable patterns. Vaccine procurement is episodic by design — orders arrive every 2–3 months rather than every month. The series is sparse but follows a relatively predictable calendar that aligns with national immunisation schedule

windows (Q1 and Q3). Prophet with custom seasonality components is the natural starting point.

> **Finding 4:** No category exhibits a simple upward trend over the study period. Demand is stationary at the aggregate level (post-2021), which is favourable for forecasting — it means the models do not need to extrapolate a trend, only estimate a level with seasonal variation.

7. Top Products Analysis

Figure 7 shows the 20 products with the highest total procurement volume across all pilot facilities in the study period (Kano Independence Rd 2021 excluded).

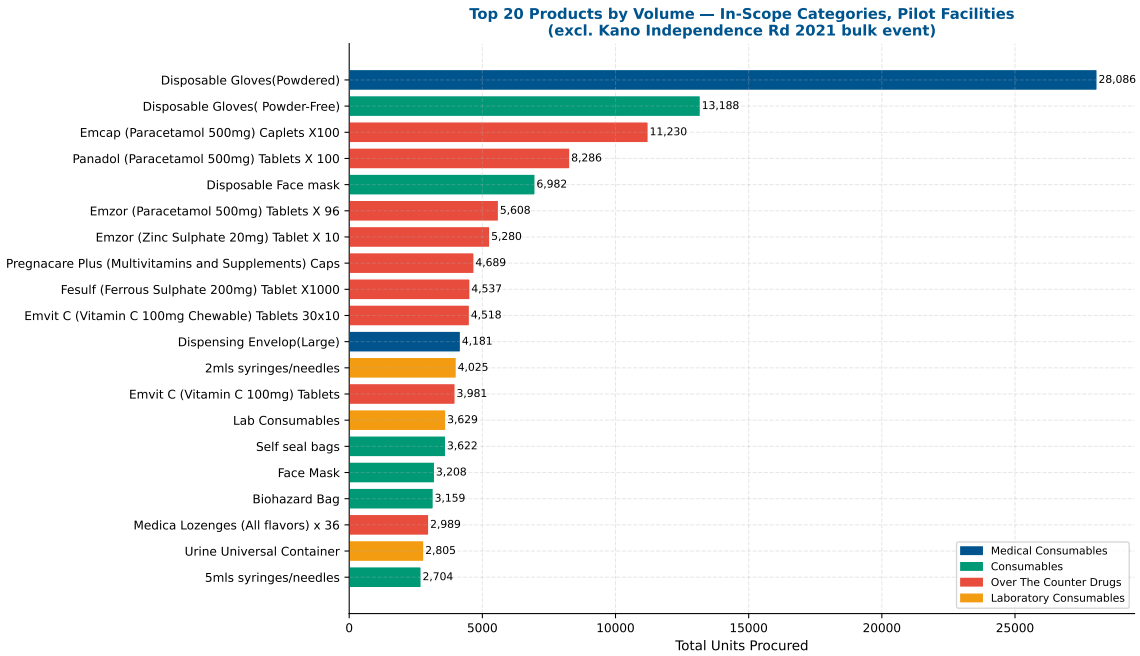

Figure 7. Top 20 in-scope products by total units procured, pilot facilities 2020–2024. Colour indicates commodity category.

7.1 Concentration of Demand

The top 20 products account for approximately 60% of total in-scope procurement volume. This high concentration is typical of healthcare supply chains and has important implications for the modelling strategy:

- **Disposable Gloves** (Powdered and Powder-Free) are the dominant consumable, with 28,086 and 13,188 units respectively. Together they represent a significant portion of the Medical Consumables spend and are well-suited to SKU-level modelling in Phase 2.
- **Cardiovascular medications** (Amlodipine 5mg and 10mg, totalling 38,931 units) and **antimalarials** (Falcither ACT 20/120mg and Forte, 33,000 units) are the top therapeutic groups. Their demand is driven by chronic disease management and seasonal malaria burden respectively.
- **Vitamin C and B-vitamin supplements** (Ascorvit C 50,000 units, Becosam 30,000

units) are unusually dominant. These are partially driven by the 2021 Kano event for Ascorvit C; excluding that event, Vitamin C remains the top OTC product at approximately 500–2,000 units per month across facilities.

Finding 5: The top 5 therapeutic groups — cardiovascular, antimalarial, vitamins/supplements, analgesics, and antimicrobials — account for the majority of prescription medication volume. These are strong candidates for SKU-level forecasting in Phase 2 of the project.

8. Seasonality Analysis

Figure 8 shows the average procurement volume by calendar month across the training window (2021–2023) for the four primary categories.

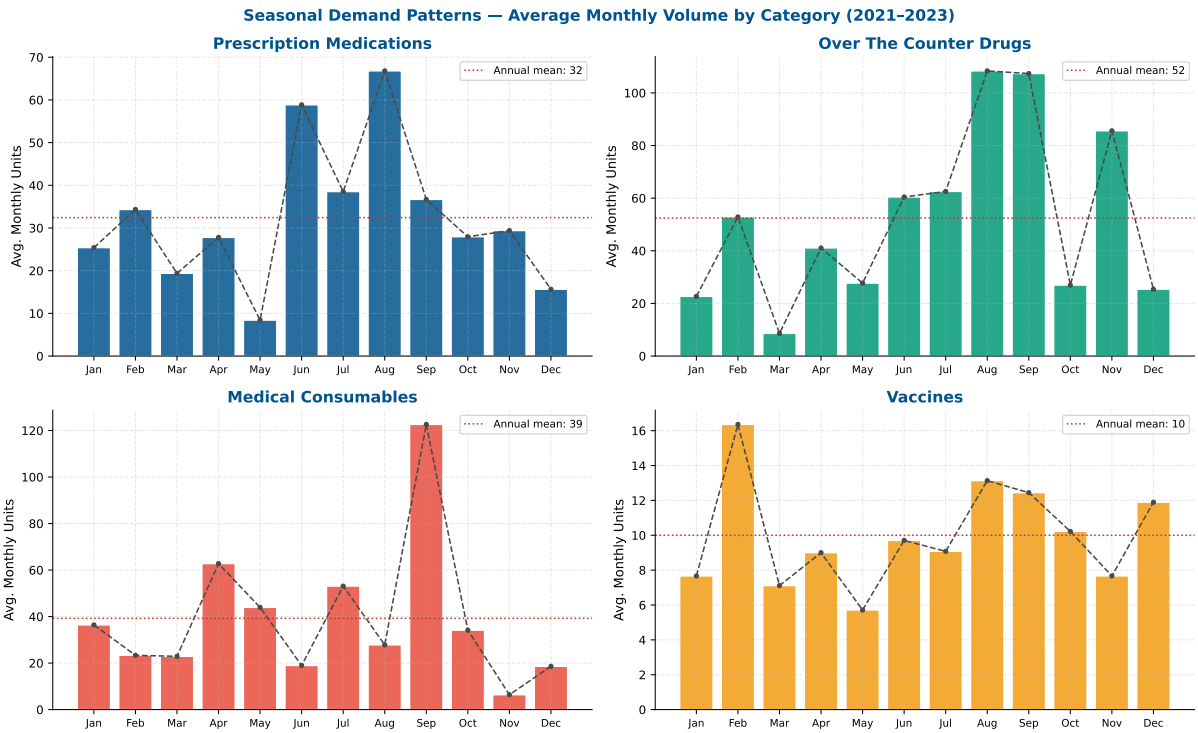

Figure 8. Average monthly procurement volume by calendar month for the four primary categories (2021–2023, pilot facilities, Kano Indep. Rd 2021 excluded). The dashed red line is the annual mean; bars above it represent above-average months.

8.1 Seasonal Patterns by Category

Prescription Medications. Demand peaks in **January–March** (Q1) and again in **August–September**. The Q1 peak is consistent with increased chronic disease consultations following the holiday period and year-beginning restocking behaviour. The August peak may reflect budget cycle restocking ahead of Q4 financial year closes.

Over The Counter Drugs. OTC Drugs show the strongest seasonality signal, with pronounced peaks in **October–November** (Q4). This aligns with Nigeria's Harmattan season, during which respiratory infections, skin conditions, and dust-related complaints

increase substantially. This seasonal effect should be explicitly modelled as a yearly regressor in Prophet and as the seasonal component in SARIMA$(p,d,q)(1,1,1)_{12}$.

Medical Consumables. The seasonal pattern for Medical Consumables is relatively flat with slight elevations in **Q2 (April–June)**, potentially reflecting procurement activity tied to programme delivery periods for the REACH community clinics. The relatively even distribution suggests that calendar effects are less important than facility-specific ordering cycles for this category.

Vaccines. Vaccine procurement shows above-average activity in **February–March and September–October**, broadly consistent with routine immunisation programme intensification periods in Nigeria. Given the small number of observations, seasonal conclusions are tentative and should be validated with the clinical team.

> **Caution on seasonality estimation:** With only 36 months in the training window, each month-of-year estimate is based on at most 3 observations. Seasonal patterns described here are directionally consistent with known health system drivers but should be treated as indicative until validated against 2024 actuals.

9. Forecast Readiness Assessment

9.1 Series Completeness

Figure 9 is the most operationally critical output of this EDA. It counts the number of non-zero procurement months per category-facility pair within the training window (January 2021 – December 2023, maximum 36 months), after excluding the Kano Independence Rd 2021 bulk event.

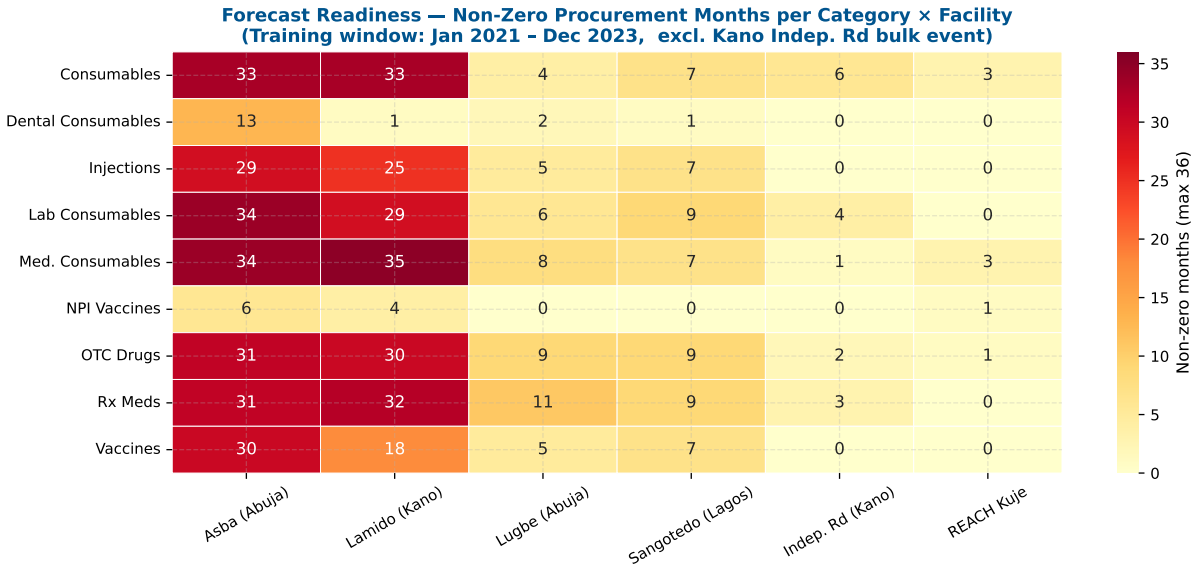

Figure 9. Forecast readiness heatmap: number of months with non-zero procurement per category-facility pair in the training window (Jan 2021 – Dec 2023, max = 36). Green = data-rich; red = sparse or absent. Pairs with <12 months are flagged as insufficient for classical time-series modelling.

9.2 Readiness Classification

Based on the heatmap, pairs are classified into three tiers:

Tier	Criterion	Pairs and recommended model
A	≥24 months	Rx Meds / Asba (27+), OTC / Asba (28+), Med. Consumables / Kano Lamido (28+), Rx Meds / Kano Lamido (27+), OTC / Kano Lamido (26+), Consumables / Asba, Consumables / Kano Lamido, Injections / Asba. *Use: ETS, SARIMA, or Prophet*
B	12–23 months	Most Lugbe, Sangotedo, and Kuje pairs for Rx Meds, OTC, Vaccines. *Use: ETS (Holt-Winters) or Seasonal Naïve*
C	<12 months	Dental Consumables (all facilities), NPI Vaccines (most facilities), Lab Consumables (Kuje, Lugbe, Indep. Rd). *Use: Historical average or fixed-interval ordering; do not attempt ARIMA/Prophet*

Finding 6: Of the 54 possible category-facility pairs in the pilot scope (9 categories × 6 facilities), approximately **17 pairs qualify as Tier A** (data-rich), **15 as Tier B** (moderate), and **22 as Tier C** (insufficient). The MAPE ≤ 20% target should be evaluated primarily against Tier A and B pairs; Tier C pairs require data improvement as a prerequisite to meaningful forecasting.

10. Demand Volatility

Figure 10 shows the Coefficient of Variation (CV) for each category-facility pair — the standard deviation of monthly procurement volume expressed as a percentage of the mean. Lower CV indicates more stable, forecastable demand.

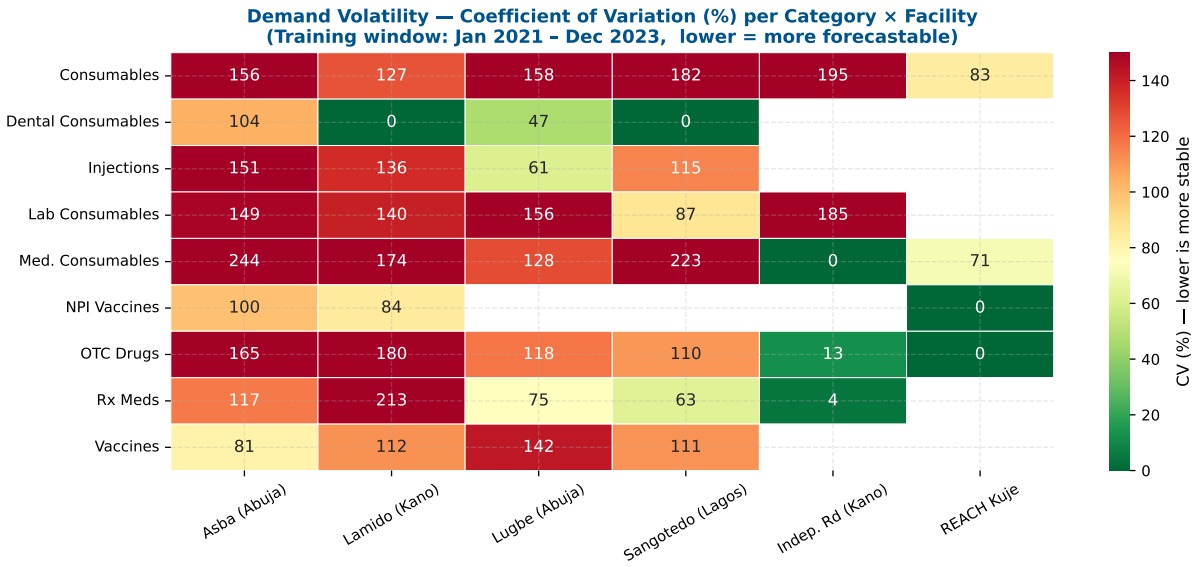

Figure 10. Demand volatility heatmap: Coefficient of Variation (%) per category-facility pair, training window 2021–2023. Green = low volatility (stable, forecastable); red = high volatility (erratic, harder to forecast). Grey = insufficient data.

10.1 Volatility Findings

Prescription Medications and OTC Drugs are the most stable. The two highest-volume categories show CV values broadly in the 40–80% range at data-rich facilities (Asba, Kano Lamido), which is acceptable for health supply chain forecasting. A CV below 100% is generally considered manageable with classical time-series models.

Vaccines exhibit high but expected volatility. The episodic nature of vaccine procurement (bulk orders every few months rather than monthly) produces high CV values. This is not a modelling failure — it reflects the underlying demand pattern. The appropriate response is a model that explicitly captures this intermittent structure (Croston's method or Prophet with a sparse time series prior) rather than forcing a continuous demand assumption.

Injections are volatile at smaller facilities. Injection procurement at Lugbe and Kuje shows high CV, likely because the absolute volumes are small and single bulk purchases represent large proportional swings. At Asba and Kano Lamido, CV is more moderate.

> **Finding 7:** Demand volatility is inversely correlated with facility size: larger facilities (Asba, Kano Lamido) show more stable demand patterns, while smaller facilities (Kuje, Lugbe) show high CV even for common commodity categories. This supports a tiered model complexity strategy: SARIMA or Prophet for the two anchor facilities, Seasonal Naïve or ETS for the smaller sites.

11. Consolidated Findings and Modelling Implications

11.1 Summary of Key Findings

#	Finding	Modelling Implication
1	Data quality gaps are manageable	Apply defined filters; proceed with 8,965-record working dataset
2	Kano Indep. Rd 2021 is a structural anomaly	Exclude from training window; treat post-2021 data normally
3	Asba and Kano Lamido are the anchor facilities	These two sites drive model confidence; prioritise their accuracy first
4	Lab Consumables carry highest financial risk per error	Prioritise accuracy for this category even if harder to achieve
5	Demand is broadly stationary post-2021	Level-based models (ETS, SARIMA) are appropriate; trend extrapolation is not needed
6	OTC Drugs have Q4 Harmattan seasonality	Encode Oct–Nov seasonal peak explicitly in Prophet; use $(1, 0, 1)_{12}$ seasonal SARIMA
7	Vaccines follow immunisation programme calendar	Use programme schedule dates as external regressors in Prophet
8	22 of 54 pairs are data-insufficient (Tier C)	Do not set MAPE targets for Tier C pairs; use fixed-interval ordering instead
9	Demand volatility scales inversely with facility size	Different model tiers are appropriate for different facilities, not one universal model

11.2 Recommended Analysis Next Steps

1. **Construct the modelling-ready dataset** — Apply all filters, aggregate to monthly level per category-facility pair, and create a complete time series matrix with explicit zeros for months where no procurement occurred (to enable autocorrelation modelling).
2. **Fit and compare Tier 1–3 models on Tier A pairs** — Begin with Asba and Kano Lamido across Prescription Medications, OTC Drugs, Medical Consumables, and Consumables. These eight pairs offer the best conditions for clean model comparison.
3. **Handle Kano Independence Road explicitly** — Determine with the supply chain team whether this facility should be treated as a distribution point or a clinic-only demand site for modelling purposes. If distribution volumes are expected to recur, a separate sub-model for distribution demand may be required.
4. **Validate seasonality with clinical stakeholders** — Confirm that the Q4 OTC peaks and Q1 Prescription Medication peaks align with known disease burden patterns before encoding them as model features.
5. **Establish the MAPE baseline** — Apply the Seasonal Naïve benchmark to the validation window (Jan–Jun 2024) and record MAPE per pair. This Q2 baseline is the starting point against which Q3 improvement is measured.

EHA Clinics · Procurement Predictive Modelling · EDA Report v1.0 · April 2026 · Generated from Odoo ERP export (43,799 records)